Bittu Kumar

MLOps Engineer | LLMOps Engineer

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PROFILE SUMMARY

MLOps and LLMOps Engineer with 2+ years of experience in designing, deploying, monitoring, and operationalizing Machine Learning and Generative AI applications. Skilled in MLOps, LLMOps, Retrieval-Augmented Generation (RAG), Kubernetes, Kubeflow, KServe, CI/CD automation, LLM evaluation, observability, and cloud-native AI infrastructure. Experienced in supporting enterprise AI solutions for the UK Water Utility industry, focusing on scalable, reliable, and production-ready ML and LLM systems.

SKILLS

LLMOps: RAG, Prompt Management, LLM Evaluation, LangSmith, DeepEval, Vector Databases

MLOps: Kubeflow, KServe, MLflow, SageMaker, Model Deployment, Model Monitoring

Cloud & DevOps: AWS, Docker, Kubernetes, GitHub Actions, ArgoCD, Terraform

Programming & Databases: Python, Bash, PostgreSQL, Qdrant

Observability: Prometheus, Grafana

PROFESSIONAL EXPERIENCE

MLOps Engineer | EXL Service

Client: UK Water Utility Industry | Apr 2024 – Present

- Designed and implemented an end-to-end MLOps workflow for customer churn prediction models, covering model training, versioning, deployment, serving, scaling, and lifecycle management.
- Built and productionized ML models with a clear separation of concerns between model logic, serving layer, and infrastructure following MLOps best practices.
- Implemented custom model serving using Flask and Gunicorn (WSGI) to understand low-level model serving before adopting managed platforms.
- Containerized ML workloads using Docker, ensuring reproducible builds and immutable model artifacts across environments.
- Deployed ML models on Kubernetes clusters using Deployments, Services, and Ingress for scalable and resilient inference.
- Implemented KServe-based model serving using standardized inference APIs and cloud-native deployment patterns.
- Integrated external model artifact storage using Amazon S3 and HTTP-based endpoints for decoupled model deployment.
- Applied SageMaker Domain, User Profiles, and IAM execution roles to align ML environments with enterprise security requirements.
- Designed IAM-based role separation for Data Scientists, ML Engineers, and MLOps Engineers within SageMaker environments.
- Applied CI/CD automation principles for model training, artifact generation, validation, and deployment readiness.
- Implemented end-to-end pipeline orchestration using Kubeflow for automated data ingestion, preprocessing, model training, evaluation, and deployment.
- Implemented monitoring for model inference performance, prediction latency, data drift, and model drift detection in production.
- Developed expertise in model reliability, deployment safety, infrastructure automation, operational scalability, and governance.

LLMOps Engineer | EXL Service

Client: UK Water Utility Industry (Thames Water) | Apr 2024 – Present

- Managed enterprise RAG pipelines powering an Operations Knowledge Copilot used across Billing, Complaints, Meter Exchange, GAP, PMP, VMD, and De-Reg business functions.
- Maintained document ingestion, chunking, embedding generation, and vector indexing workflows for SOPs, policies, operational procedures, process guides, and knowledge base content.
- Administered vector databases and optimized retrieval quality through metadata filtering, embedding management, and search tuning strategies.
- Implemented LLM evaluation frameworks to measure answer relevance, faithfulness, context precision, context recall, and hallucination rates.
- Established observability and monitoring for production LLM applications, tracking latency, token consumption, retrieval quality, availability, and operational costs.
- Managed prompt lifecycle, prompt versioning, testing, and release management processes across enterprise AI applications.
- Automated CI/CD workflows for prompt validation, RAG evaluation, testing, and production deployments.
- Managed embedding lifecycle, vector database maintenance, and retrieval optimization to improve operational knowledge discovery and response accuracy.
- Implemented AI guardrails, governance controls, and human-in-the-loop review workflows to improve reliability, compliance, and response quality.
- Supported deployment, scaling, monitoring, and operational maintenance of containerized LLM applications on Kubernetes environments.
- Managed feedback collection, evaluation datasets, and continuous improvement processes to enhance AI system performance.

EDUCATION

Bachelor of Computer Applications (BCA)
2020 – 2023

AWARDS & RECOGNITIONS

- Received Spot Award for outstanding contribution and performance at EXL Services.
- Recognized for delivering high-quality solutions and supporting critical business initiatives for the UK Water Utility client.
- Appreciated for driving automation, operational excellence, and continuous improvement across AI/ML initiatives.